

Propiedades de las involuciones del disco unidad

Lema 1. Sea $w \in \mathbb{D}$ y τ_w la involución del disco unidad asociada a w , entonces

- | | |
|--|--|
| (1) $z \in \mathbb{D} \iff \tau_w(z) < 1.$ | (3) $\tau_w \in \text{Aut}(\mathbb{D}).$ |
| (2) $z \in \mathbb{T} \iff \tau_w(z) = 1.$ | (4) $\tau_w \circ \tau_w = \text{id}.$ |

Demostración:

- (1) Usaremos que $\forall z_1, z_2 \in \mathbb{C} : |z_1 + z_2|^2 = |z_1|^2 + 2\Re(\overline{z_1}z_2) + |z_2|^2$. Sea $z \in \mathbb{C}$, entonces

$$\begin{aligned}
 |\tau_w(z)| < 1 &\iff \left| \frac{w - z}{1 - \overline{w}z} \right|^2 < 1 \iff |w|^2 - 2\Re(\overline{w}z) + |z|^2 < 1 - 2\Re(\overline{w}z) + |w|^2|z|^2 \\
 &\iff 1 + |w|^2|z|^2 - |w|^2 - |z|^2 > 0 \iff (1 - |w|^2)(1 - |z|^2) > 0 \\
 &\stackrel{|w| < 1}{\iff} 1 - |z|^2 > 0 \iff |z| < 1 \iff z \in \mathbb{D}.
 \end{aligned}$$

- (2) El mismo cálculo de (1) nos da que $|\tau_w(z)| = 1 \iff z \in \mathbb{T}$.

- (3) Por (1), $\tau_w(\mathbb{D}) = \mathbb{D}$ y por (2), $\tau_w(\mathbb{T}) = \mathbb{T}$. Como además τ_w es una transformación de Möbius, es holomorfa e inyectiva, por lo que $\tau_w \in \text{Aut}(\mathbb{D})$.

$$(4) \quad \tau_w \circ \tau_w(z) = \tau_w \left(\frac{w - z}{1 - \overline{w}z} \right) = \frac{w - \frac{w - z}{1 - \overline{w}z}}{1 - \overline{w} \cdot \frac{w - z}{1 - \overline{w}z}} = \frac{w(1 - \overline{w}z) - (w - z)}{(1 - \overline{w}z) - \overline{w}(w - z)} = \frac{z - |w|^2 z}{1 - |w|^2} = z.$$

■

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.